

Do LLMs Understand Language? Part 2

LIN 313 Language and Computers
UT Austin Fall 2025
Instructor: Gabriella Chronis

Overview

How do we evaluate LLMs understanding?

- simulation: simulating the future
- abstraction: can LLMs generalize to out of distribution test domains?

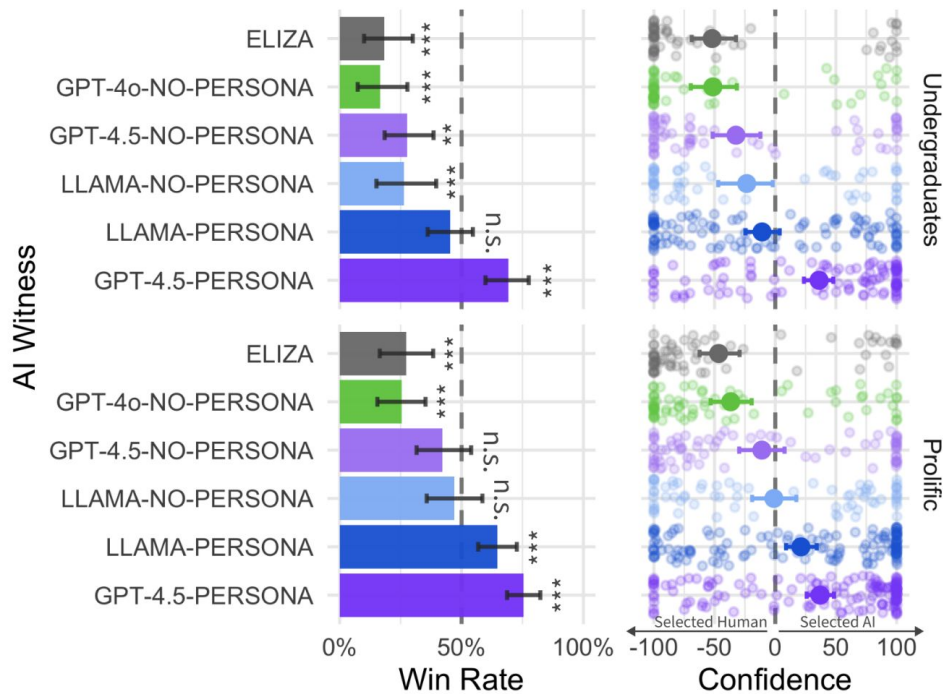
How do we give LLMs reasoning capabilities?

- Retrieval-augmented generation (RAG)
 - reduces hallucination
- Chain of Thought (CoT)
 - fewer logical errors

Can LLMs think?

- they pass the Turing test
- they get good test scores

LLMs Pass the TT????



LLMs have reason???

Exam results (ordered by GPT-3.5 performance)

Estimated percentile lower bound (among test takers)

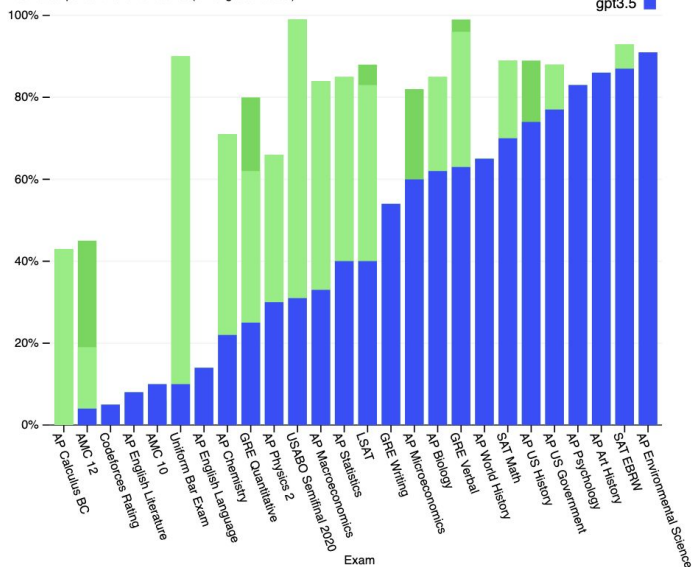


Figure 4. GPT performance on academic and professional exams. In each case, we simulate the conditions and scoring of the real exam. Exams are ordered from low to high based on GPT-3.5 performance. GPT-4 outperforms GPT-3.5 on most exams tested. To be conservative we report the lower end of the range of percentiles, but this creates some artifacts on the AP exams which have very wide scoring bins. For example although GPT-4 attains the highest possible score on AP Biology (5/5), this is only shown in the plot as 85th percentile because 15 percent of test-takers achieve that score.

	GPT-4 Evaluated few-shot	GPT-3.5 Evaluated few-shot	LM SOTA Best external LM evaluated few-shot	SOTA Best external model (incl. benchmark-specific tuning)
MMLU [49] Multiple-choice questions in 57 subjects (professional & academic)	86.4% 5-shot	70.0% 5-shot	70.7% 5-shot U-PaLM [50]	75.2% 5-shot Flan-PaLM [51]
HellaSwag [52] Commonsense reasoning around everyday events	95.3% 10-shot	85.5% 10-shot	84.2% LLaMA (validation set) [28]	85.6 ALUM [53]
AI2 Reasoning Challenge (ARC) [54] Grade-school multiple choice science questions. Challenge-set.	96.3% 25-shot	85.2% 25-shot	85.2% 8-shot PaLM [55]	86.5% ST-MOE [18]
WinoGrande [56] Commonsense reasoning around pronoun resolution	87.5% 5-shot	81.6% 5-shot	85.1% 5-shot PaLM [3]	85.1% 5-shot PaLM [3]
HumanEval [43] Python coding tasks	67.0% 0-shot	48.1% 0-shot	26.2% 0-shot PaLM [3]	65.8% CodeT+ GPT-3.5 [57]
DROP [58] (F1 score) Reading comprehension & arithmetic.	80.9 3-shot	64.1 3-shot	70.8 1-shot PaLM [3]	88.4 QDGAT [59]
GSM-8K [60] Grade-school mathematics questions	92.0%* 5-shot chain-of-thought	57.1% 5-shot	58.8% 8-shot Minerva [61]	87.3% Chinchilla + SFT+ORM-RL, ORM reranking [62]

Table 2. Performance of GPT-4 on academic benchmarks. We compare GPT-4 alongside the best SOTA (with benchmark-specific training) and the best SOTA for an LM evaluated few-shot. GPT-4 outperforms existing LMs on all benchmarks, and beats SOTA with benchmark-specific training on all datasets except DROP. For each task we report GPT-4’s performance along with the few-shot method used to evaluate. For GSM-8K, we included part of the training set in the GPT-4 pre-training mix (see Appendix E), and we use chain-of-thought prompting [11] when evaluating. For multiple-choice questions, we present all answers (ABCD) to the model and ask it to choose the letter of the answer, similarly to how a human would solve such a problem.

CopyCat (Mitchell)

Analogy problems

abc -> abd, abcc -> ?

If abc changes to abd, what does ace change to?

Q: If abc changes to abd, what does pqr change to?

A: pqs

Q: If abc changes to abd, what does pqrs change to?

<https://aiguide.substack.com/p/on-analogy-making-in-large-language>

LLMs and Copycat

- Many "probing" studies that publish results like, "yes, wow, LLMs can do X complicated task that we associate with reasoning and abstraction

Article | Published: 31 July 2023

Emergent analogical reasoning in large language models

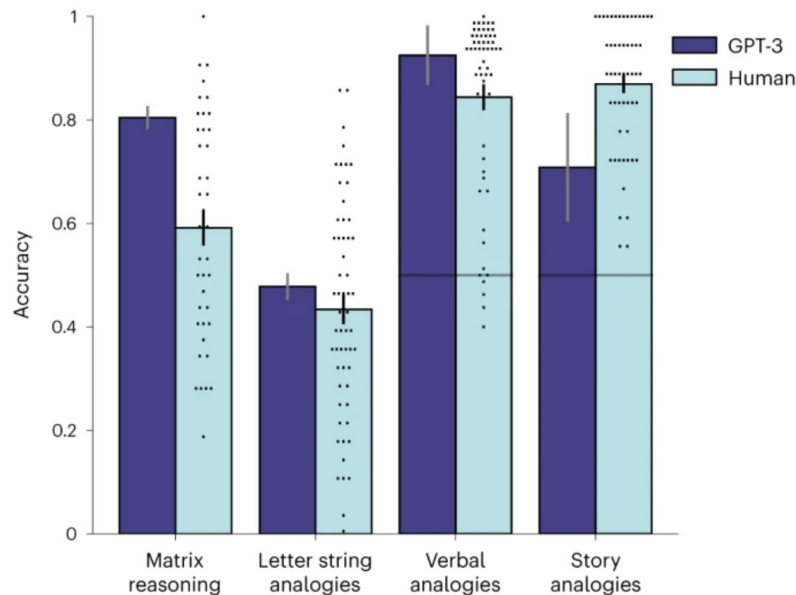
Taylor Webb , Keith J. Holyoak & Hongjing Lu

Nature Human Behaviour 7, 1526–1541 (2023) | [Cite this article](#)

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 Matters Arising to this article was published on 04 June 2025

Fig. 1: Summary of results.



Can LLMs think?

- They pass the Turing test
 - maybe this test tells us more about people than it does about LLMS
- They get good test scores
 - GPT's test results look pretty good! If it can do PhD level math and chemistry problems, it must have abstract reasoning abilities, right?
 - WRONG
 - **data contamination:** overlap between training data and test data.
 - The model might just be memorizing the right answer, or using shallow **heuristics** rather than true symbolic reasoning.

To evaluate understanding, we have to get creative.

But first... What is Understanding?

Mitchell: Three domains of understanding

- intuitive physics
- intuitive psychology
- intuitive biology

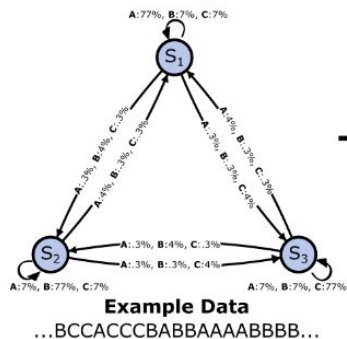
Two cognitive-scientific ideas of understanding

- **Simulation:** predicting possible futures
 - Barsalou (1999): situated concepts: Understanding and reasoning involve mentally simulating relevant perceptual, motor, and introspective experiences
- **Abstraction and analogy**
 - Doug Hofstadter (2007): "the perception of a common essence between two things"
 - Bruno Latour (1987): abstractions are not immaterial, but material accomplishments that require maintenance and support. Abstract concepts don't exist "above" or separate from the material world; they're held in place by concrete practices and connections.



LLMs Simulate the (entire) future

Data Generating Structure



Theoretical Prediction



Transformers Represent Belief State Geometry in their Residual Stream

Adam S. Shai
Simplex
PIBBSS*

Sarah E. Marzen
Department of Natural Sciences
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PIBBSS

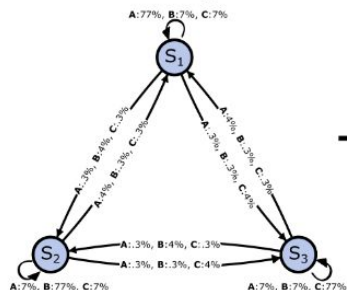
Alexander Gietelink Oldenziel
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BITS

- trained small transformer models on artificial "language" data generated by a stochastic markov model
- trained model to predict just the next token
- demonstrated that the model keep track of different possibilities for what the future might hold

LLMs Simulate the (entire) future

Data Generating Structure

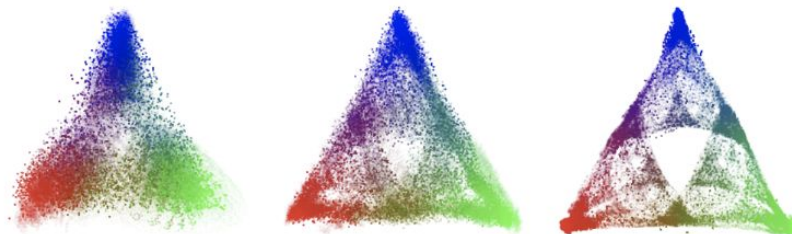


Example Data
...BCCACCCBABBAAABBBB...

Theoretical Prediction



Residual Stream Geometry



Training

Transformers Represent Belief State Geometry in their Residual Stream

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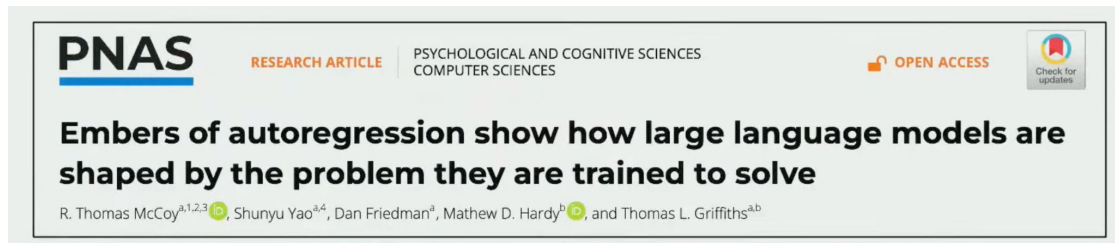
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Testing for Abstraction (Thomas McCoy et al.)

Q: How are we supposed to know if the model actually learned an abstract rule or just learned shallow heuristics like memorizing a bunch of examples (a.k.a. spurious correlations)??

A: Look at how it generalizes on out-of-domain (Ood) questions



<https://www.pnas.org/doi/pdf/10.1073/pnas.2322420121>

Experiment: Article Swapping

Task: Swap each article (a, an, the) with the previous word

Article swapping

Swap each article (*a*, *an*, or *the*) with the word before it.

Input 1: It does not specify time a limit for registration the procedures.

Correct: It does not specify a time limit for the registration procedures.

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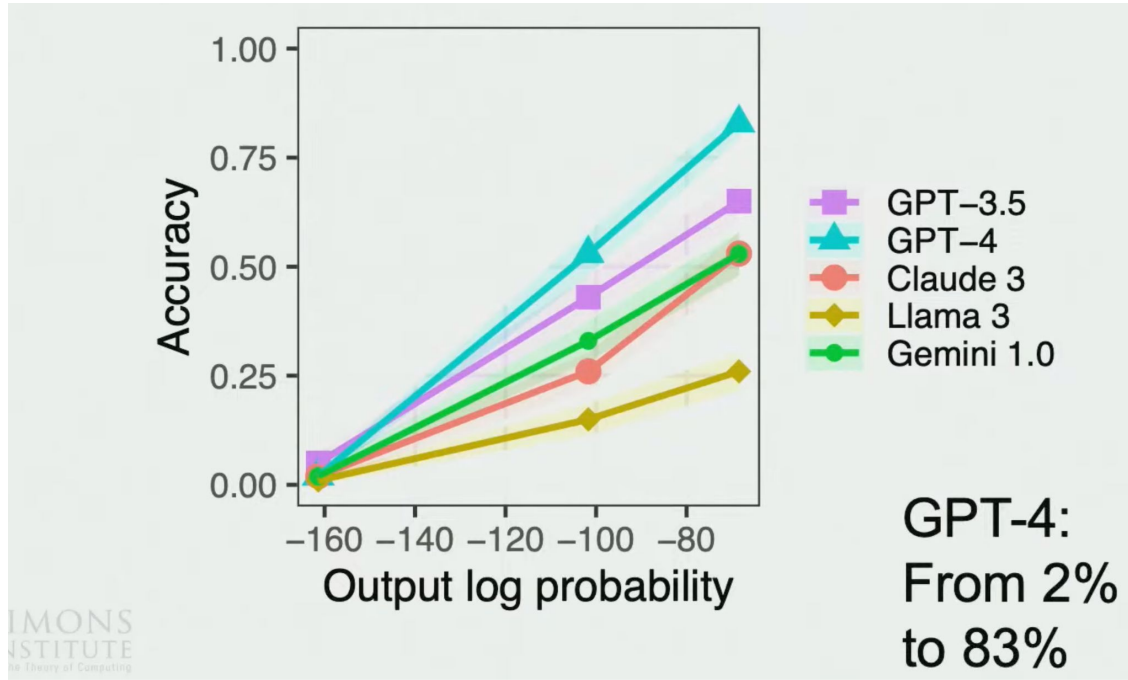
Input 2: It few with it to lying take the get just a hands would kinds.

Correct: It few with it to lying the take get a just hands would kinds.

✗ **GPT-4:** It flew with a few kinds to take the lying just to get the hands.

Article Swapping: Probability to Sensitivity

The model does a lot better at the task if the input string is high probability (aka a grammatical sentence)

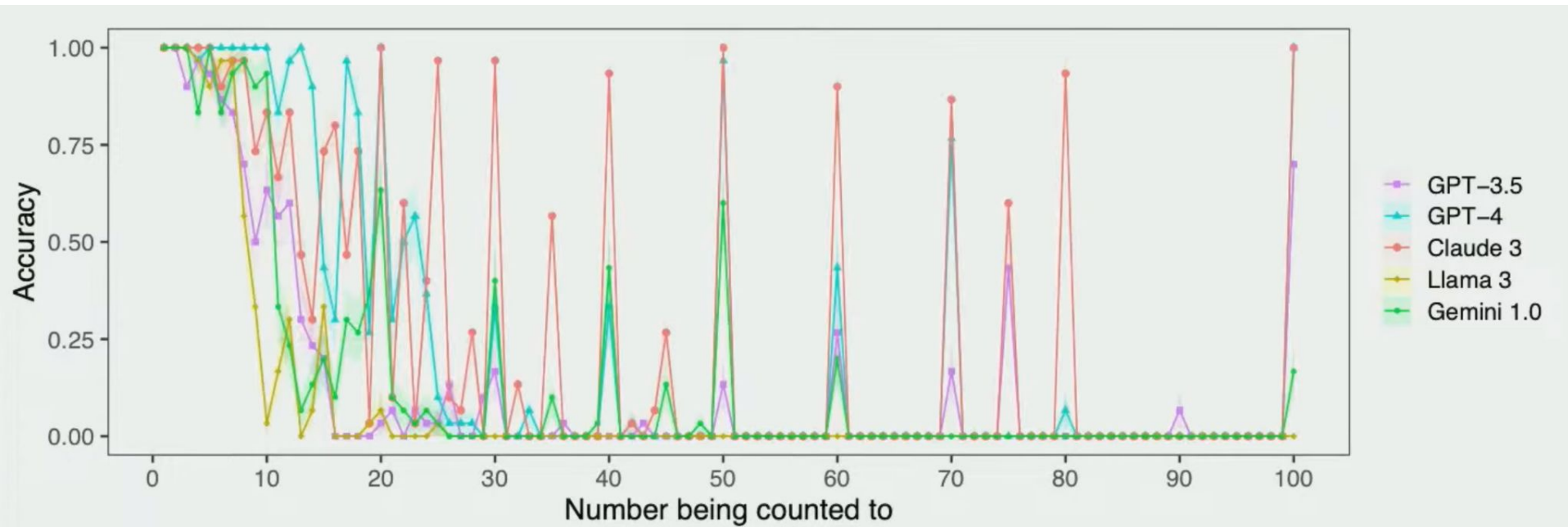


Experiment: Counting words

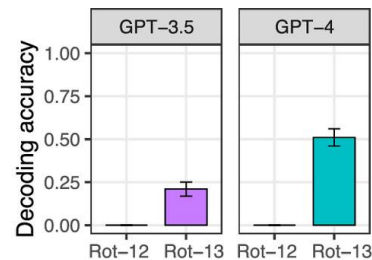
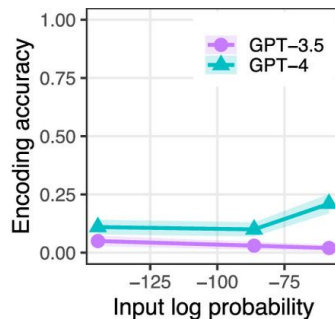
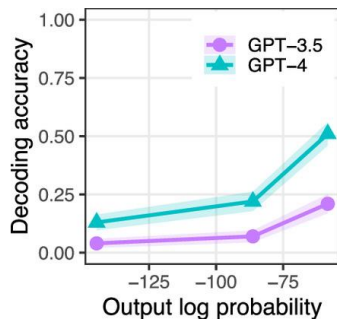
- The task:
 - How many words are in this list? "lively news exhibit sleep"

Experiment: Counting words

- The task:
 - How many words are in this list? "lively news exhibit sleep"



Experiment: Caesar Cipher



Note: In Internet text, rot-13 is about 175 times more common than rot-12.

Shift cipher: Output probability

Rot-13 decoding: high-probability output

Input: Svefg, fur whfg cbfgrq gb ure Vafgntenz fgbel.

Correct: First, she just posted to her Instagram story.

✓ **GPT-4:** First, she just posted to her Instagram story.

Rot-13 decoding: medium-probability output

Input: Fbeel, Naanguba jevgrf gb bhe Pbclevtug Hfref.

Correct: Sorry, Annathon writes to our Copyright Users.

✗ **GPT-4:** Sorry, Annabeth writes to our Prophetic Users.

Shift cipher: Input probability

Rot-13 encoding: high-probability input

Input: In a word, everything has been complicated there.

Correct: Va n jbeq, rirelguvat unf orra pbzcyvpngrq gurur.

✓ **GPT-4:** Va n jbeq, rirelguvat unf orra pbzcyvpngrq gurur.

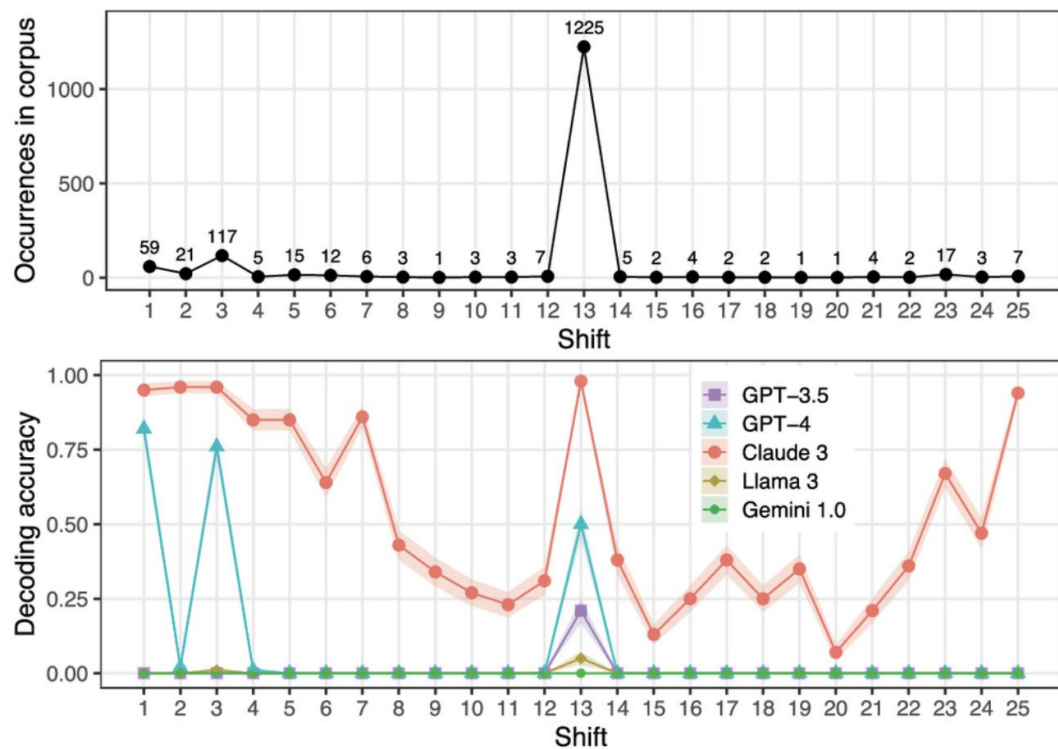
Rot-13 encoding: medium-probability input

Input: In a word, governance has been frustrating daily.

Correct: Va n jbeq, tbireanapr unf orra sehfgengvat qnyvl.

✗ **GPT-4:** Va n jbeq, tbinapr unf orra sehfgevat qnyl1.

Experiment: Caesar Cipher



General Results

Overall, the more probable the input or output string is, the better the model does.

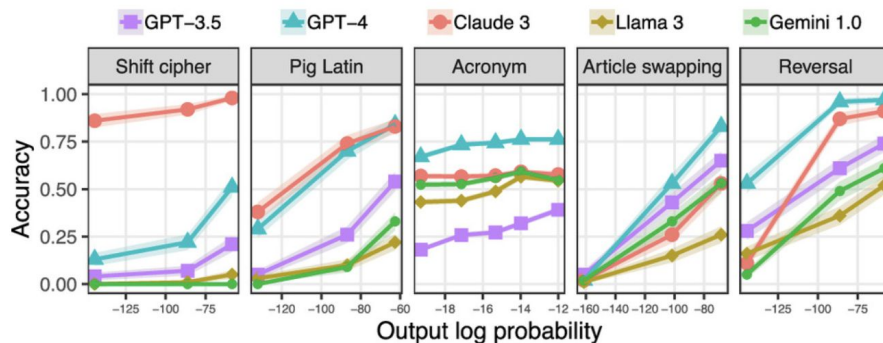


Fig. 4. The effect of output probability on model accuracy across five tasks. The intervals around the lines show one SE.

Three methods for enhancing LLMs

So far we have talked about

1. LLMs: the base language model (a decoder-only transformer model)
 - a. trained on web text to predict the next word
 - b. will finish any given text where it leaves off
2. RLHF: reinforcement learning with human feedback
 - a. trained to respond like a good chatbot
 - b. will *respond* to a text

There are three more methods that get us to the current day (e.g. GPT-5)

- Agentic AI
- Retrieval-Augmented Generation (RAG)
- Chain of Thought (CoT)

But first, the System Prompt

prompt: the string of text that gets passed to the model.

system prompt: a special, high-level instruction that sets the rules, role, goals, and constraints for the model's behavior throughout a conversation.

- acts the model's contextual operating system:
- tells the model how to behave, what to prioritize, and what persona or capabilities to adopt when generating responses.

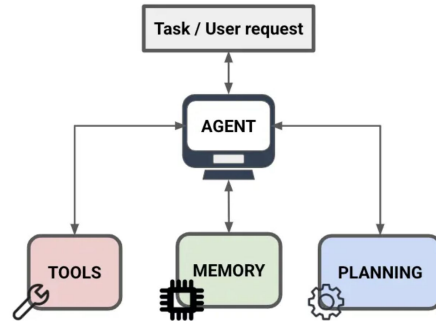
GPT5's system prompt:

https://github.com/jujumilk3/leaked-system-prompts/blob/main/openai-chatgpt5_20250808.md

Agentic AI: LLMs do a lot more than just generating text these days

They write code, they generate documents, they buy plane tickets, they update calendars....

They also remember you! How does that work without more model training?



How Do LLMs Cite their sources?



Is there an overuse or overreliance on AI?



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◆ AI Overview

Yes, there are significant concerns about the overuse and overreliance on AI, leading to potential issues like the erosion of critical thinking, skill degradation, and a failure to scrutinize incorrect AI outputs. Overreliance can manifest as blindly accepting AI recommendations (algorithmic susceptibility) or even confidently ignoring incorrect AI advice (algorithmic aversion). Experts suggest balancing AI efficiency with human oversight and focusing on AI literacy to mitigate these risks. 

Risks of overreliance on AI

- **Erosion of critical thinking:** Excessive use of AI for tasks can lead to a decline in independent problem-solving and a reduced ability to perform higher-order thinking.
- **Skill degradation:** Cognitive skills can deteriorate as people lose opportunities to practice and develop them, leading to a "brain drain" in areas like creativity and analysis.
- **Acceptance of errors:** Overreliance can cause users to accept and act on incorrect AI outputs without proper scrutiny, leading to erroneous decisions in critical fields like medicine or finance.
- **Lack of transparency and accountability:** When individuals don't understand how an AI reached a conclusion, it creates a "black box" problem, reducing accountability and critical engagement.
- **Inappropriate trust:** The mere knowledge that advice is coming from an AI can cause people to over-trust it, even when it contradicts other available information or their own judgment. 

AI Overreliance Is a Problem. Are Explanations a Solution? | Stanford HAI

Mar 12, 2023 — AI overreliance is when humans accept an AI system's recommended decision...

 Stanford HAI 



Trust and reliance on AI — An experimental study on the ...

In a domain-independent, incentivized, and interactive behavioral experiment, we find that...

 ScienceDirect.com 



AI Tools in Society: Impacts on Cognitive Offloading and the ... - MDPI

Jan 2, 2025 — One concern is the risk of cognitive dependence, where individuals...

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How Do LLMs Cite their sources?

Where do these links come from?

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Answer: Retrieval-Augmented Generation (RAG)

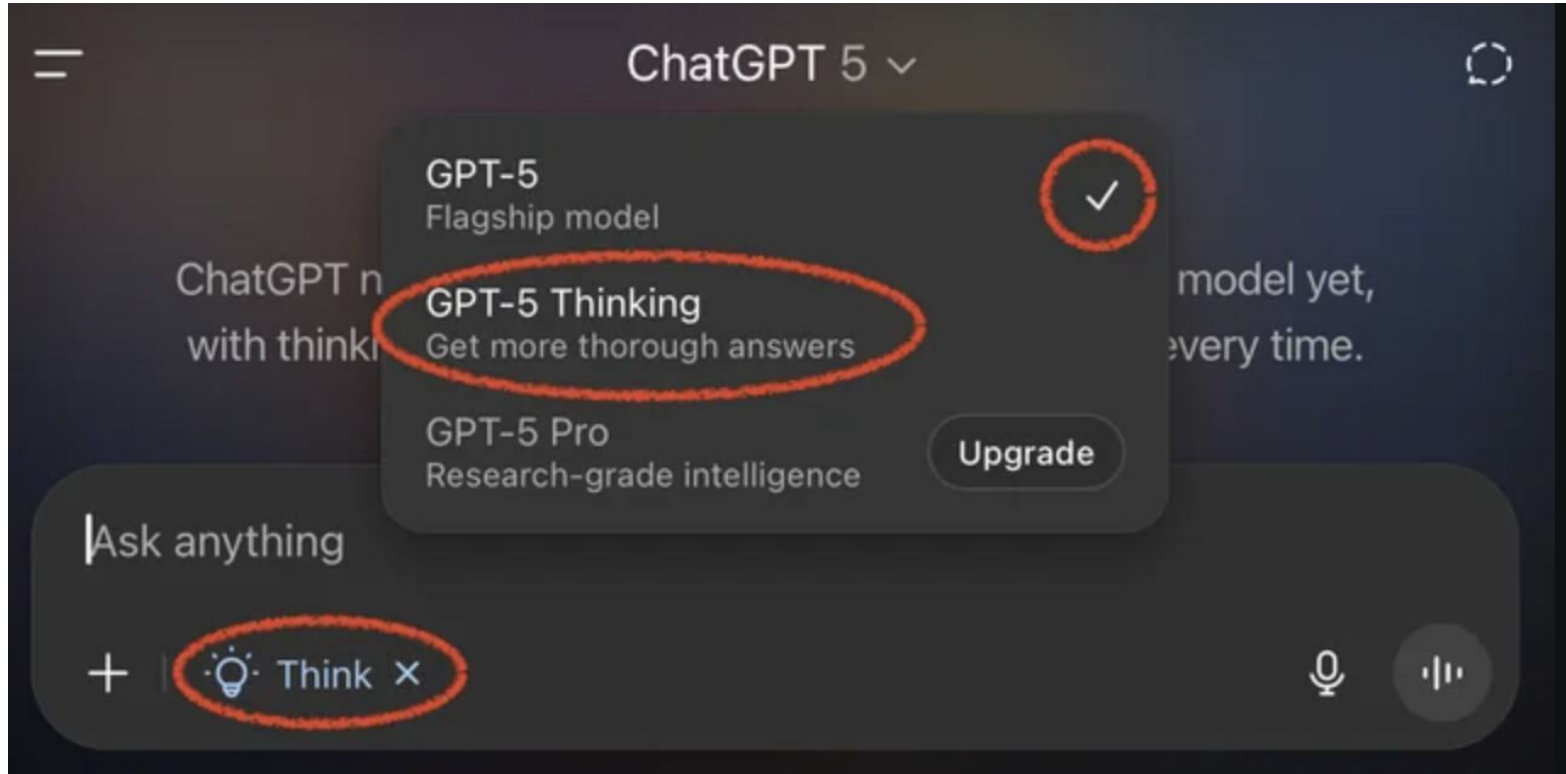
1. Information extraction: regular web search returns replies

- a. uses a newer version of the document retrieval method on HW5 - vectors to represent documents
- b. get text of the **top N web pages**

2. Prompt generation

- a. **system prompt** (not exact):
 - i. "You are a search overview assistant. Your job is to answer the user's query using information from provided documents. Use specific information from the documents and cite them when you do as [source 1] [source 2] etc. Your answer should be concise, no more than 500 words."
- b. all of this gets passed to the LLM as a big string of text
- c. new prompt = **system prompt** + "user: " + your prompt + **text from source 1** + . . . + **text from source N** + "response: "

What is a "thinking" model?



Chain of Thought

[Submitted on 28 Jan 2022 (v1), last revised 10 Jan 2023 (this version, v6)]

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, Denny Zhou

Ask the model to :

1. show your work
2. justify your answer
3. revise if necessary

Standard Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain of Thought Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Regular Prompting vs CoT (Wei et al.)

CoT on Symbolic Reasoning Tasks

Tasks:

- **Last letter concatenation.** This task asks the model to concatenate the last letters of words in a name (e.g., “Amy Brown” → “yn”).
- **Coin flip.** This task asks the model to answer whether a coin is still heads up after people either flip or don’t flip the coin (e.g., “A coin is heads up. Phoebe flips the coin. Osvaldo does not flip the coin. Is the coin still heads up?” → “no”).

CoT prompting shows massive improvement over plain prompting.

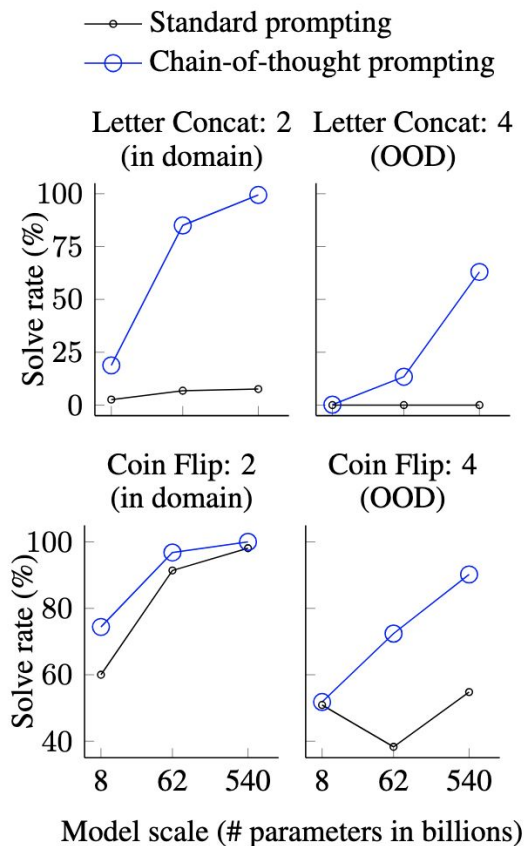


Figure 8: Using chain-of-thought prompting facilitates generalization to longer sequences in two symbolic reasoning tasks.

LLMs and Copycat

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Article | Published: 31 July 2023

Emergent analogical reasoning in large language models

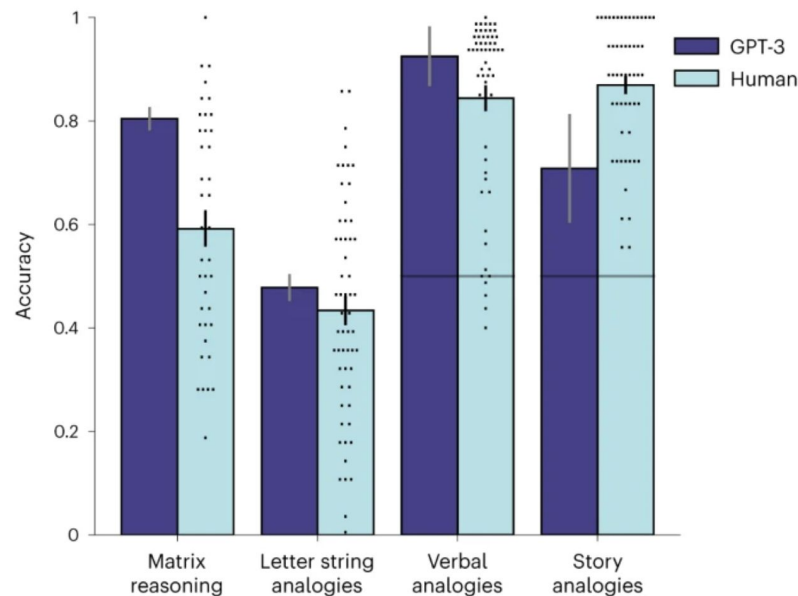
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Fig. 1: Summary of results.



Can Large Language Models generalize analogy solving like children can? (Stevenson et al. 2024)

- Ask LLMs to solve analogies roman Roman alphabet, Greek alphabet, and Symbols



$ab \rightarrow ac$

$gh \rightarrow$

a b c d e f g h i j k l m n o p q
r s t u v w x y z



(a) Latin

$\alpha\beta \rightarrow \alpha\gamma$

$\zeta\eta \rightarrow$

$\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi$



(b) Greek

$* @ \rightarrow * \%$

$\sim \$ \rightarrow$

* @ % ! ^ # ~ \$ { ? :]



(c) Symbol

Can Large Language Models generalize analogy solving like children can? (Stevenson et al. 2024)

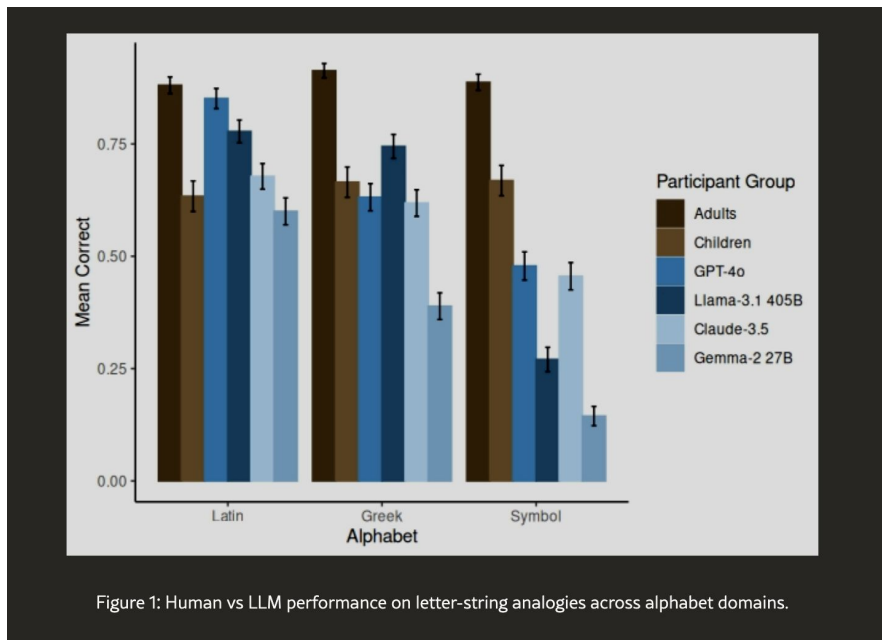


Figure 1: Human vs LLM performance on letter-string analogies across alphabet domains.